

Scaling Report — ACT, relative-EE, strawberry picking

Question: How does ACT validation loss scale with training-data size? **Setup:** 3 nested train subsets — **100 / 500 / 1012 episodes** — identical ACT policy (UMI relative-EE, rot6d, chunk30, 52M params), identical optimizer (AdamW 1e-5, batch 8), shared **100-episode validation set**. Full method in `EXPERIMENT.md`.

The 100-ep and 500-ep runs were **stopped early at ~1.93–1.95M steps** (out of 2.5M) once the result was unambiguous — both had overfit their best val by ~1.5M steps, so no signal was lost. Best-val is the primary metric throughout.

Headline result

Validation loss follows a clean **power law in dataset size**:

$$L(D) \approx 0.27 \cdot D^{-0.26} \quad (D = \text{training episodes, fit on best val})$$

episodes D	best val loss	(fit predicts)	final val loss	best-val step
100	0.0833	0.0840	0.0870	380k
500	0.0572	0.0556	0.0625	70k
1012	0.0456	0.0465	0.0480	180k

Power-law predictions match all three points within $\pm 3\%$ (effectively exact with 3 points $\rightarrow$ exponent $b \approx 0.26$, episodes; ≈ 0.29 if D is measured in frames). In practical terms: **$\sim 10\times$ more data (100 $\rightarrow$ 1012 ep) roughly halves val loss (0.083 $\rightarrow$ 0.046), or each doubling of data cuts val loss by $\sim 16\%$ ($2^{-0.26} \approx 0.84$).**

Figures

Figure 1 — scaling (2 $\times$ 2). (A) val loss vs steps, (B) train loss vs steps, (C) val loss vs dataset size — log-log with power-law fit, (D) generalization gap vs steps.

Figure 2 — loss components (validation). L1 (left) and KLD (right) vs steps.

wandb (all grouped in `biorobotlab/lerobot_scaling_ee`): 100ep · 500ep · 1012ep (mirror).

Findings

1. Data scales — power law, with strong diminishing returns. Monotonic, near-perfect log-log linearity across 100 $\rightarrow$ 1012 ep. But the shallow ex-

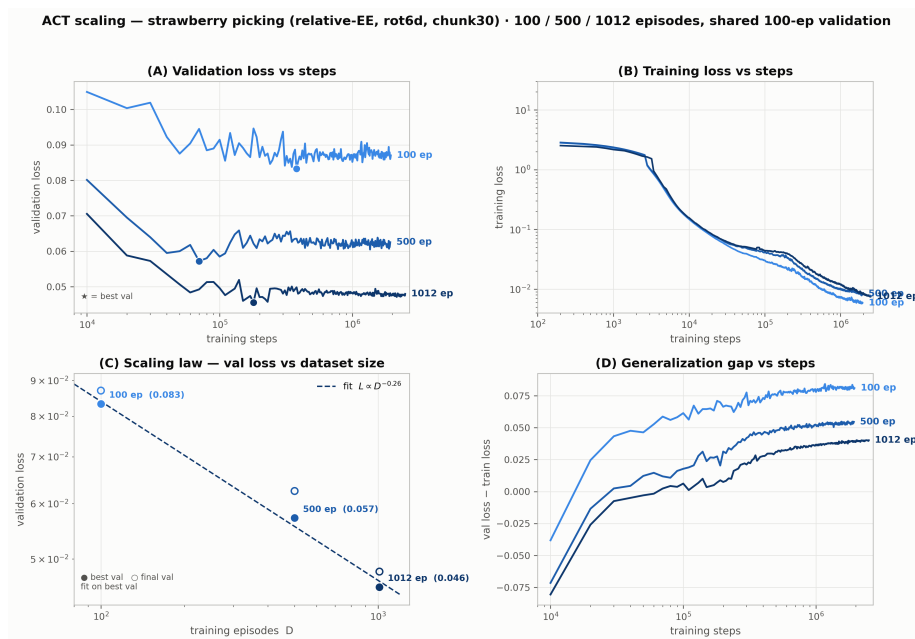


Figure 1: Figure 1 — scaling

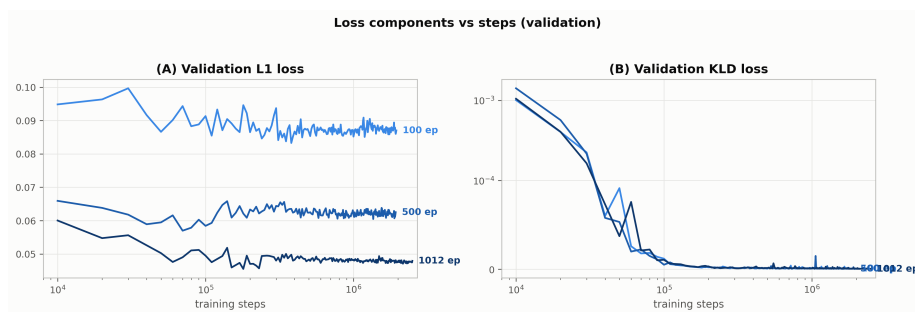


Figure 2: Figure 2 — loss components

ponent (0.26) means **loss is expensive to push down**: extrapolating the fit, reaching **L=0.04** needs **~1.6k episodes** and **L=0.03** needs **~5k** — i.e. roughly 5–15× more data than the current 1012 set for a modest further reduction. Data alone has mostly played out at this scale; architectural / representation gains likely dominate beyond here.

2. Everything overfits early — the 2.5M-step budget is ~10–30× too large. Best validation arrives at **70k (500-ep)**, **180k (1012-ep)**, **380k (100-ep)** — then val loss plateaus and slowly drifts up while train loss keeps dropping (memorization). For these data scales, **~300–500k steps** captures the achievable val loss; training to 2.5M only adds overfit. Implication: for deployment, early-stop on val (~best-val step).

3. Generalization gap collapses with data. At each run’s best-val step, gap (val-train) = **0.070 (100-ep)** → **0.012 (500-ep)** → **0.007 (1012-ep)**. The 100-ep model memorizes its small set (train 0.006 vs val 0.083); the 1012-ep model generalizes (train 0.008 vs val 0.046). This is the mechanism behind the scaling law: more data shrinks the gap, which shrinks val loss. (*Panel D.*)

4. Loss is L1-dominated; KLD vanishes (VAE posterior collapse). KLD → ~0 within the first ~50k steps for every run (the ACT VAE posterior collapses to the prior), so essentially **all of val loss is L1** (reconstruction). KLD is not a useful discriminator across data sizes here. (*Fig 2.*)

Caveats

- **Leading-prefix subsets:** 100/500 are the *first* N episodes by session order, so at small N early recording sessions are over-represented (100-ep ≈ 117 frames/ep vs ≈87 for the full set). Fine for a nested sweep; the 100-ep point may be slightly optimistic/pessimistic depending on early-session difficulty.
- **3 data points** → the exponent is a rough estimate (log-linear fit on 3 points). Treat $b \approx 0.26$ as ± 0.05 .
- **Single seed per point** — no error bars; val curves are noisy (± 0.005 band) but the inter-run gaps (0.083 / 0.057 / 0.046) far exceed the noise.
- 100/500 runs **stopped at ~1.94M**, not 2.5M (best-val fully captured earlier; see table).
- Normalization stats recomputed per subset (correct for fair comparison; means the MIN_MAX bounds differ slightly across runs).

Reproduce

- Runs launched via the parent directory’s `train_relative_ee_processor.py` (see `EXPERIMENT.md` for the exact config & subsets).
- Figures regenerated by `make_figures.py`, which reads per-run history CSVs (`_f100,500,1012_history.csv`) produced by pulling each run’s metrics from wandb (`run.history(...)`) or parsing the run console logs.

Power-law fit is a log-linear `np.polyfit` on best-val points.

Bottom line

Val loss scales as a power law $L \propto D^{-0.26}$ over 100→1012 demos, halving per decade of data. The model overfits within a few hundred k steps regardless of data size, so the practical lever past 1k demos is no longer “more data” — returns have diminished enough that representation/architecture changes should outperform further data collection. For this setup, **~1012 episodes + early-stop near its best val (~180k steps) is the efficient operating point.**